

Embedding torus into hierarchical crossed cubes *

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Abstract

Mesh and torus are popular interconnection topology and many distributed-memory computers use mesh and torus as the basis of their communication network. And now, many parallel algorithms with mesh and torus structure have been developed. If one network can embed a mesh or torus network, the algorithms with mesh-structured or torus-structured can also be used in this network. Hierarchical crossed cubes(HCC) have been proposed with many interesting properties such as low degree, logarithmic diameter, simple algorithm. In this paper, we propose an algorithm to embed a $2^{n-1} \times 2^n$ torus into a $HCC_{1,n}$ with dilation 4.

interconnection structure of HCC has many advantages such as lower degree, smaller diameter and maximum fault tolerance in the same number of vertices. The problems of hamiltonian cycle, path cover and property of pancyclicity on hierarchical crossed cube have been investigated in [2, 3, 14, 17]. In this paper, we will propose an algorithm to embed a $2^{n-1} \times 2^n$ torus into a $HCC_{1,n}$ with dilation 4.

The rest of this paper is organized as follows. We present definition of hypercube, crossed cube and hierarchical crossed cube in Section 2 and introduce reflected edge label sequence to generate two kinds of cycles. Then, we prove that we can embed a torus into $HCC_{1,n}$ in Section 4. Conclusion is given in the final section.

1 Introduction

The mesh and torus are two important interconnection networks. The parallel algorithms of mesh-structured and torus-structured have been developed and are used in many distributed-memory computers [6, 15].

An n -dimensional crossed cube, CQ_n , is an important variation of hypercube Q_n . CQ_n has 2^n vertices and $n2^{n-1}$ links, the same as hypercube Q_n . However, the diameter of crossed cube is almost the half of hypercube. Embedding a k -dimensional mesh into crossed cube has been proposed in recent years[4, 5]. Furthermore, Tsai et al. proposed a linear algorithm to embed a mesh and torus into CQ_n [13].

Recently, hierarchical interconnection networks have attracted lots of attention[9, 16]. A new hierarchical interconnection network, the hierarchical crossed cube $HCC_{k,n}$, was proposed in [8]. The topology is suitable for the design of massive parallel system with thousands of processors. The

2 Preliminaries

Interconnection network plays a major role in distributed systems and parallel computing. A topology of an interconnection network is conveniently represented with an undirected graph $G(V, E)$, where vertices in V represents the nodes and the edges in E represents the links between two vertices. A *path* is a sequence of adjacent vertices written as $\langle u_0, u_1, \dots, u_m \rangle$, and all the vertices $u_0, u_1, \dots, u_m$ are distinct. If $u_0 = u_m$, then the path is called a *cycle*. A cycle traversing each vertex of G exactly once is a *Hamiltonian cycle*. We define $d_G(u, v)$ as the distance between two vertices u and v in G .

Given two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$, an embedding from guest graph G_1 to host graph G_2 is an injective mapping $\psi : V(G_1) \rightarrow V(G_2)$. If the network of host graph G_2 can be embedded with the guest graph G_1 , the algorithms designed for guest graph can also be used in the host graph network. The performance of embedding problem can be measured by *dilation*. The dilation of embedding ψ is defined like this : $\max\{d_{G_2}(\psi(u), \psi(v))\}$ and $(u, v) \in E(G_1)$.

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An n -dimensional hypercube, denoted by Q_n , has 2^n nodes. We use n -bit strings to denote the vertices set of Q_n . There is an edge joining two vertices if and only if their labels differ in exactly one bit position.

The n -dimensional crossed cube CQ_n is a variation of hypercube. Similar to the n -dimensional hypercube, its vertex set is $\{0,1\}^n$. Let $R = \{(00,00), (10,10), (01,11), (11,01)\}$ and let two digit binary strings $x = x_1x_0$ and $y = y_1y_0$ are pair related, as $x \sim y$, if and only if $(x,y) \in R$. The n -dimensional crossed cube CQ_n is defined recursively and is built from two disjoint copies of an $(n-1)$ -dimensional crossed cubes, CQ_{n-1}^0 and CQ_{n-1}^1 , where the name of any vertex in CQ_{n-1}^i is that of the corresponding vertex from CQ_{n-1} prefixed with the binary value i , $i = 0,1$. The vertex $0x_{n-2}x_{n-3} \cdots x_1x_0$ of CQ_{n-1}^0 is joined to the vertex $1y_{n-2}y_{n-3} \cdots y_1y_0$ of CQ_{n-1}^1 if and only if

- (1) $x_{n-2} = y_{n-2}$, if n is even;
- (2) $(x_{2i+1}x_{2i}, y_{2i+1}y_{2i}) \in R$, for $0 \leq i < \lceil \frac{n}{2} \rceil - 1$

The hierarchical crossed cube $HCC_{1,n}$ has vertex set $\{0,1\}^{1+2^n}$. Each vertex of $HCC_{1,n}$ is written as (u,v,w) , where $u \in \{0,1\}$ and $v,w \in \{0,1\}^n$. The set of edges of $HCC_{1,n}$ is partitioned into two sets, E_{int} and E_{ext} . The set E_{int} is referred to as the set of *internal* edges, whilst the set E_{ext} is referred to as the set of *external* edges. For convenience, let E_0 denote the label of the external edges on $HCC_{1,n}$.

- (1) $E_{int} = \{((u,v,w), (u,v,w')), (w,w') \text{ is an edge of } CQ_n\}$
- (2) $E_{ext} = \{((u,v,w), (\bar{u}, w, v), \bar{u} = 1 - u)\}$

Figure 1 and 2 illustrate the $HCC_{1,1}$ and $HCC_{1,2}$ respectively.

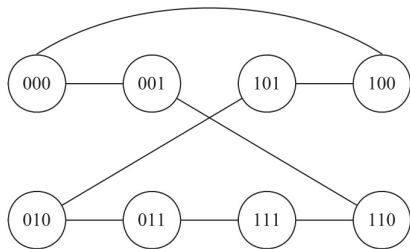


Figure 1: $HCC_{1,1}$

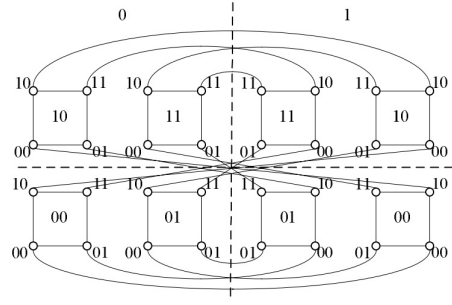


Figure 2: $HCC_{1,2}$

3 Interleaved cycle

For any two adjacent vertices u, v , the edge $(u, v) \in E(Q_n)$ (respectively, $E(CQ_n)$) is labeled by i if $u_i \neq v_i$ and $u_j = v_j$ for $i+1 \leq j \leq n-1$ and v is the i -th dimensional neighbour of u . The edge (u, v) can also be denoted by $u[i]v$.

A path in Q_n (respectively, CQ_n) could be specified by the source vertex and an ordered sequence of edge labels (or dimensions), for example, the path detailed as having the source vertex 000 and then following the edge sequence 1,2,1 (also denoted by $[1,2,1]$) is actually the path $\langle 000, 010, 110, 100 \rangle$ (also denoted $000[1, 2, 1]100$ or $\langle 000[1, 2, 1]100 \rangle$).

Let $L = [d_1, d_2, \dots, d_m]$ be an edge label sequence in Q_n (respectively, CQ_n) where two successive labels are not identical and $d_i \in Z_n$, $Z_n = \{0, 1, \dots, n-1\}$, for $1 \leq i \leq m$.

Herein, we focus on a special edge label sequence termed the reflected edge label sequence generated by a systematic method. Let $\pi_l = (d_1, d_2, \dots, d_l)$, $1 \leq l \leq n$, be a permutation with l elements over on Z_n , and let $\pi_l(i) = (d_1, d_2, \dots, d_i)$. The *reflected edge label sequence*, defined by π_l , is generated recursively as follows:

$$\begin{aligned} RL_{\pi_l(1)} &= d_1, \\ RL_{\pi_l(i)} &= RL_{\pi_l(i-1)}, d_i, RL_{\pi_l(i-1)}, 2 \leq i \leq l, \text{ and} \\ RL_{\pi_l} &= RL_{\pi_l(l)} \end{aligned}$$

As a result, the RL_{π_l} defined by an arbitrary permutation π_l on Z_n is a symmetry edge label sequence in Q_n (respectively, CQ_n). For convenience, let $C_{\pi_l} = [RL_{\pi_l}, d_l]$. We called an edge label sequence L a Hamiltonian cycle pattern (HCP for short) of Q_n (respectively, CQ_n) if for each u in a Q_n (respectively, CQ_n), $u[L]u$ forms a Hamiltonian cycle of a Q_n (respectively, CQ_n). Moreover, π_n is termed a Hamiltonian cycle pattern generator (HCPG for short) if S_{π_n} is a HCP.

Lemma 1 [1] For $n \geq 3$, let $\pi_n = (d_1, \dots, d_{n-1}, d_n)$ be a permutation with n elements over Z_n . Then, π_n is a Hamiltonian cycle pattern generator of a CQ_n if and only if
 (1) $\min\{d_{n-1}, d_n\}$ is odd, or
 (2) $|d_n - d_{n-1}| = 1$.

Lemma 2 [12] A $HCC_{1,n}$ can be decomposed into 2^{n-1} vertex disjoint cycles of length 2^{n+2} .

Let π_n be a HCPG of CQ_n and let S_{π_n} be the reflected edge label sequence generated by π_n . Let $C_{\pi_n}^i$ be the edge label sequence generated by rotating left S_{π_n} with $2i$ positions. Let $C_{\pi_n}^i[j]$ denote the j -th element in $C_{\pi_n}^i$, $0 \leq j \leq 2^n - 1$. For instance, for $\pi_3 = \{0, 1, 2\}$, then $C_{\pi_3} = (0, 1, 0, 2, 0, 1, 0, 2)$, $C_{\pi_3}^1 = (0, 2, 0, 1, 0, 2, 0, 1)$, and $C_{\pi_3}^1[3] = 1$.

In $HCC_{1,n}$, an *interleaved cycle* is a cycle $C = \langle w_0, w_1, \dots, w_{2^{n+1}-1}, w_0 \rangle$ that edges are alternately internal edges and external edges, which contains exactly one internal edge in each corresponding CQ_n . We now give an algorithm to generate the sequences for interleaved cycles as Algorithm 1.

Algorithm 1 Generating sequences of interleaved cycles

Input: a HCPG π_n of CQ_n
Output: 2^{n-1} sequences, denoted by IS^i $0 \leq i \leq 2^{n-1} - 1$, of interleaved cycles.
 1: Generate C_{π_n} with π_n
 2: **for** $0 \leq i \leq 2^{n-1} - 1$ **do**
 3: Rotate left S_{π_n} with $2i$ positions.
 4: **for** $0 \leq j \leq 2^n - 1$ **do**
 5: $IS^i[4j] = C_{\pi_n}^i[j]$.
 6: $IS^i[4j + 1] = E_0$.
 7: $IS^i[4j + 2] = C_{\pi_n}^i[j]$.
 8: $IS^i[4j + 3] = E_0$.
 9: **end for**
 10: **end for**

Algorithm 2 Generating interleaved cycles [12]

Input: $C = \omega(C_{\pi_n}, \mathbf{w})$
 1: **for** $0 \leq i \leq 2^{n-1} - 1$ **do**
 2: **for** $0 \leq j \leq 2^n - 1$ **do**
 3: $IC^i[4j] = (0, \mathbf{w}_j, \mathbf{w}_{2i+j})$.
 4: $IC^i[4j + 1] = (0, \mathbf{w}_j, \mathbf{w}_{2i+j+1})$.
 5: $IC^i[4j + 2] = (0, \mathbf{w}_{2i+j+1}, \mathbf{w}_j)$.
 6: $IC^i[4j + 3] = (0, \mathbf{w}_{2i+j+1}, \mathbf{w}_{j+1})$.
 7: **end for**
 8: **end for**

For a HCPG π_n and any vertex $\mathbf{w}$ of a CQ_n , we can construct a Hamiltonian cycle as $C = \omega(C_{\pi_n}, \mathbf{w})$. Let $\mathbf{w}_0 = \mathbf{w}$ and $\mathbf{w}_i$ denote i -th vertex on C . Moreover, let $x_{2i} = (0, 0^n, \mathbf{w}_{2i})$ be a vertex of $HCC_{1,n}$ and let $IC^i = \omega(IS^i, x_{2i})$. Given a start vertex x_i , we can use IS^i to generate a corresponding interleaved cycle. It is not difficult to verify that the interleaved cycles are consistent to the result of Algorithm 2. Hence, we have Lemmas 3 and 4. Figure 3 presents four disjoint interleaved cycles of a $HCC_{1,3}$, produced by Algorithm 2.

Lemma 3 [12] For a HCPG π_n and the $2i$ -th vertex $\mathbf{w}_{2i}$ of $\omega(C_{\pi_n}, 0^n)$, $IC^i = \omega(IS^i, x_{2i})$ forms an interleaved cycle of a $HCC_{1,n}$ where $x_{2i} = (0, 0^n, \mathbf{w}_{2i})$.

Lemma 4 [12] Given a HCPG π_n and a vertex $\mathbf{w}$ of CQ_n , let $\mathbf{w}_{2i}$ and $\mathbf{w}_{2j}$ be $2i$ -th and $2j$ -th vertices of $\omega(C_{\pi_n}, \mathbf{w})$, $0 \leq i, j \leq 2^{n-1} - 1$ and $i \neq j$. Then IC^i and IC^j of Algorithm 2 are disjoint cycles.



Figure 3: Visualizing four disjoint interleaved cycles of a $HCC_{1,3}$

Now we have known how to construct 2^{n-1} disjoint cycles on a $HCC_{1,n}$. Then, our purpose is embedding a torus into a $HCC_{1,n}$, so we need to find out another edge label sequence that will

vertically cross with the interleaved cycles. Let vertical edge label sequence $VS_{\pi_n}^i$ be the edge label sequence generated by rotating left S_{π_n} with i position and let $VS_{\pi_n}^i[j]$ denote the j -th element in $VS_{\pi_n}^i$. Then we give an algorithm to generate the vertical edge label sequence as following.

Algorithm 3 Generating vertical sequences

- 1: Generate C_{π_n} with π_n
 - 2: **for** $0 \leq i \leq 2^n - 1$ **do**
 - 3: Rotate left C_{π_n} with i position.
 - 4: $VS^i = C_{\pi_n}^i$.
 - 5: **end for**
-

4 Mapping torus into HCC

Notation 1 An $i \times j$ torus M can be denoted by an $i \times j$ matrix

$$M_{i,j} = \begin{pmatrix} x_{0,0} & x_{0,1} & \cdots & x_{0,j-1} \\ x_{1,0} & x_{1,1} & \cdots & x_{1,j-1} \\ \vdots & \vdots & \ddots & \vdots \\ x_{i-1,0} & x_{i-1,1} & \cdots & x_{i-1,j-1} \end{pmatrix},$$

where $V(M) = \{x_{p,q} \mid 0 \leq p \leq i-1, \text{ and } 0 \leq q \leq j-1\}$, $(x_{p,q}, x_{p,q+1}) \in E(M)$ (“+” represent modulo i addition) for $0 \leq p \leq i-1$ and $0 \leq q \leq j-1$, and $(x_{r,s}, x_{r+1,s}) \in E(M)$ (“+” represent modulo j addition) for $0 \leq r \leq i-1$ and $0 \leq s \leq j-1$.

Theorem 1 There is a $2^{n-1} \times 2^n$ torus in a $HCC_{1,n}$ with dilation 4.

Proof. For a HCPG π_n and a vertex $\mathbf{w}$ of a CQ_n , we can construct a Hamiltonian cycle as $C = \omega(C_{\pi_n}, \mathbf{w})$. Let $\mathbf{w}_0 = \mathbf{w}$ and $\mathbf{w}_i$ denote i -th vertex on C . Without loss of generality, let vertex $x_{0,0}$ be $(0, \mathbf{w}_0, \mathbf{w}_0)$. According to the Algorithm 2, we can construct 2^{n-1} disjoint cycles like the format $\langle x_{i,0}, x_{i,1}, \dots, x_{i,2^{n-1}-1} \rangle$, $0 \leq i \leq 2^{n-1} - 1$ in a $HCC_{1,n}$.

By Algorithm 2, $x_{i,4j} = (0, \mathbf{w}_j, \mathbf{w}_{2i+j})$, $0 \leq i \leq 2^{n-1} - 1$ and $0 \leq j \leq 2^n - 1$. Note, $\langle x_{0,4j}, x_{1,4j}, \dots, x_{2^{n-1}-1,4j} \rangle$ belong to a cycle of $CQ_n(0, \mathbf{w}_j)$. Moreover, it is not difficult to verify that $\langle x_{0,4j}, x_{1,4j}, \dots, x_{2^{n-1}-1,4j} \rangle$, for $0 \leq j \leq 2^n - 1$, can be generated by Algorithm 3 with π_n and the start vertex $x_{0,4j}$. We know $d_{HCC_{1,n}}(x_{i,4j}, x_{i+1,4j}) = 2$ (“+” represent modulo 2^{n-1} addition) and $d_{HCC_{1,n}}(x_{i,4j}, x_{i,4(j+1)}) = 4$. Hence, there is a $2^{n-1} \times 2^n$ torus in a $HCC_{1,n}$

and the dilation is 4.

For example, let $x_{0,0} = (0, \mathbf{w}_0, \mathbf{w}_0)$, then a $2^2 \times 2^3$ torus mapping to $HCC_{1,3}$ can be formed as the following matrix.

$$\begin{pmatrix} (0, \mathbf{w}_0, \mathbf{w}_0) & (0, \mathbf{w}_1, \mathbf{w}_1) & (0, \mathbf{w}_2, \mathbf{w}_2) & (0, \mathbf{w}_3, \mathbf{w}_3) & (0, \mathbf{w}_4, \mathbf{w}_4) & (0, \mathbf{w}_5, \mathbf{w}_5) & (0, \mathbf{w}_6, \mathbf{w}_6) & (0, \mathbf{w}_7, \mathbf{w}_7) \\ (0, \mathbf{w}_0, \mathbf{w}_2) & (0, \mathbf{w}_1, \mathbf{w}_3) & (0, \mathbf{w}_2, \mathbf{w}_4) & (0, \mathbf{w}_3, \mathbf{w}_5) & (0, \mathbf{w}_4, \mathbf{w}_6) & (0, \mathbf{w}_5, \mathbf{w}_7) & (0, \mathbf{w}_6, \mathbf{w}_0) & (0, \mathbf{w}_7, \mathbf{w}_1) \\ (0, \mathbf{w}_0, \mathbf{w}_4) & (0, \mathbf{w}_1, \mathbf{w}_5) & (0, \mathbf{w}_2, \mathbf{w}_6) & (0, \mathbf{w}_3, \mathbf{w}_7) & (0, \mathbf{w}_4, \mathbf{w}_0) & (0, \mathbf{w}_5, \mathbf{w}_1) & (0, \mathbf{w}_6, \mathbf{w}_2) & (0, \mathbf{w}_7, \mathbf{w}_3) \\ (0, \mathbf{w}_0, \mathbf{w}_6) & (0, \mathbf{w}_1, \mathbf{w}_7) & (0, \mathbf{w}_2, \mathbf{w}_0) & (0, \mathbf{w}_3, \mathbf{w}_1) & (0, \mathbf{w}_4, \mathbf{w}_2) & (0, \mathbf{w}_5, \mathbf{w}_3) & (0, \mathbf{w}_6, \mathbf{w}_4) & (0, \mathbf{w}_7, \mathbf{w}_5) \end{pmatrix}$$

Figure 4: A $2^2 \times 2^3$ torus mapping to $HCC_{1,3}$

5 Conclusion

In this paper, we use two edge label sequences to construct a $2^{n-1} \times 2^n$ torus on a $HCC_{1,n}$ with dilation 4. So the mesh-structured and tours-structured algorithms can also be applied in a $HCC_{1,n}$. In the future, we will investigate more applications about edge label sequences on $HCC_{k,n}$.

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